

Kenny Ballou

Assistant Professor

+1.208.720.1419
kb@devnulllabs.io
kennyballou.com
kennyballou
in kennyballou
0000-0002-6032-474X

Education

- 2020 – 2024 **Ph.D. Computing, Computer Science**, *Boise State University*, Boise, Idaho
Dissertation Advisor: Dr. Elena Sherman
Dissertation: Leveraging Incrementality of Data-Flow Analysis to Improve Analysis Over Weakly-Relational Abstract Domains
- 2010 – 2014 **B.S., Applied Mathematics**, *Boise State University*, Boise, Idaho
Minor in Computer Science

Publications

Conference Proceedings

- Aug 2025 ***muRelBench: MicroBenchmarking for Zonotope Domains***, Kenny Ballou and Elena Sherman, In: Software Engineering of Emerging Technologies
DOI 10.1007/978-3-032-08977-9_35
- Oct 2023 ***Minimally Comparing Relational Abstract Domains***, Kenny Ballou and Elena Sherman, In: Automatic Technologies for Verification and Analysis
DOI: 10.1007/978-3-031-45332-8_8
- Jul 2023 ***Identifying Minimal Changes in the Zone Abstract Domain***, Kenny Ballou and Elena Sherman, In: Theoretical Aspects of Software Engineering
DOI: 10.1007/978-3-031-35257-7_13
- May 2022 ***Incremental Transitive Closure for Zonal Abstract Domain***, Kenny Ballou and Elena Sherman, In: NASA Formal Methods
DOI: 10.1007/978-3-031-06773-0_43

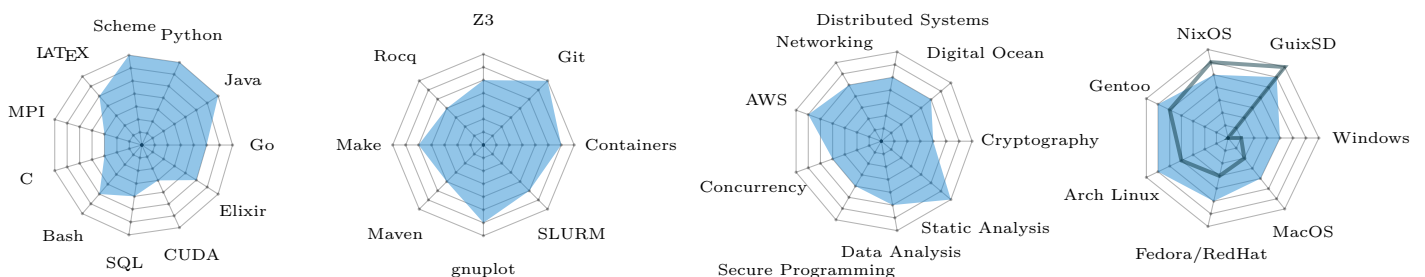
Software

- Aug 2025 ***StaticIcedTea***, Kenny Ballou and Elena Sherman, GitHub/Zenodo
DOI: 10.5281/zenodo.16693791
- Aug 2025 ***muRelBench 1.0***, Kenny Ballou and Elena Sherman, GitHub/Zenodo
DOI: 10.5281/zenodo.16694742

Books and Chapters

- Aug 2024 ***Leveraging Incrementality of Data-flow Program Analysis to Improve Analysis Over Weakly-Relational Abstract Domains***, Kenny Ballou
DOI: 10.18122/td.2260.boisestate
- Jan 2016 ***Learning Elixir***, Kenny Ballou

Technical Skills



Experience

- Aug 2024 – **Assistant Professor of Computer Science and Software Engineering**, *California State Present University, San Marcos*, San Marcos, California
- Aug 2020 – **Graduate Research Assistant**, *Boise State University*, Boise, Idaho
- Aug 2024 Research Focus: Numerical Abstract Interpretation, Static Program Analysis, Software Engineering
- Industry Employment

-
- Mar 2017 – **Site Reliability Engineer**, *HomeCU*, Boise, Idaho
- Aug 2020 Developed and maintained Amazon Web Services (AWS) infrastructure and container environments using infrastructure as code.
- Jan 2016 – **Software Engineer**, *zData*, Boise, Idaho
- Jan 2017 Developed and maintained a data platform for managed services.
- Jul 2015 – **Software Engineer**, *Clearwater Analytics*, Boise, Idaho
- Dec 2015 Developed and maintained a suite of web applications for Asset Managers.
- Jan 2015 – **Data Solution Architect**, *Alpine Data Labs*, Remote
- Jul 2015 Developed and maintained an SDK for remote installations and diagnostics. Provided technical support to sales and account managers.

Talks

- March 2024 **Leveraging Incrementality of DFA to Reason About Weakly-Relational Abstract Domains**, *Computing Colloquium*, Boise State University
- Oct 2023 **Minimally Comparing Relational Abstract Domains**, *Automated Technologies for Verification and Analysis*, Singapore
- Jul 2023 **Identifying Minimal Changes in the Zone Abstract Domain**, *Theoretical Aspects of Software Engineering*, Bristol, UK
- May 2022 **Incremental Transitive Closure for Zonal Abstract Domain**, *NASA Formal Methods*, Pasadena, CA
- Mar 2017 **Learning Git in Reverse**, *Boise Code Camp*, Boise, ID
- Elixir/Erlang Hot Loading Code**, *Boise Code Camp*, Boise, ID
- Mar 2016 **Learning Git in Reverse**, *Boise Code Camp*, Boise, ID
- Functional Programming with Elixir**, *Boise Code Camp*, Boise, ID

Teaching

- Spring 2026 **Programming Languages**, California State University San Marcos
- Spring 2026 **Software Architecture**, California State University San Marcos
- Spring 2026 **Senior Project II**, California State University San Marcos
- Fall 2025 **Data Structures and Algorithms**, California State University San Marcos
- Fall 2025 **Introduction to Software Engineering**, California State University San Marcos
- Fall 2025 **Introduction to Software Engineering Practices**, California State University San Marcos
- Fall 2025 **Senior Project I**, California State University San Marcos
- Spring 2025 **Programming Languages**, California State University San Marcos
- Spring 2025 **Game Programming**, California State University San Marcos
- Spring 2025 **Software Architecture**, California State University San Marcos
- Fall 2024 **Data Structures and Algorithms**, California State University San Marcos
- Fall 2024 **Introduction to Software Engineering**, California State University San Marcos
- Spring 2023 **Theory of Computation**, *Guest Lecture*, Boise State University
- Introduction to Finite Automata
- Spring 2022 **Theory of Computation**, *Guest Lecture*, Boise State University
- Regular Languages and Operations of DFA

- Fall 2022 **Static Program Analysis**, *Guest Lecture*, Boise State University
Introduction to Numerical Abstract Domains
- Fall 2013 **Discrete Mathematics**, *Lab Assistant*, Boise State University
- Spring 2013 **Android App Development**, *Teaching Assistant*, Boise State University
Gave guest lecture on Git
- Fall 2011 **Computer Science I and II**, *Lab Assistant*, Boise State University

Professional Service

- SEET 2026 Short Research Papers Program Chair
- CSCSU 2026 Program Committee Member
- SEET 2025 Session Chair
- CSCSU 2025 Program Committee Member
- PLDI 2022 Student Volunteer Session Chair

Honors and Awards

- 2025 CSTEM Faculty/Student Collaboration Award - Engineering
- 2013 Student Research Initiative Fellowship Recipient